

Jingbin (Justin) Lin

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EDUCATION

University of California - San Diego

Sep 2025 - Present

Master of Science, Computer Engineering (EC79)

California

- **Coursework:** Low-power VLSI Implementation for Machine Learning, Digital Communication Circuits I, Microwave Circuits, VLSI Digit Sys Algorithms & Arch, VLSI Integrated Circuits and Systems Design

McMaster University

Sep 2020 - Jun 2025

Bachelor of Engineering and Management, Electrical Engineering and Management

Canada

- **GPA:** 3.5
- **Achievements:** McMaster Honour Award, Dr. Rudolf De Buda Scholarship, Dean's Honour List
- **Coursework:** Microelectronic (CMOS Technology), Photonic Device & Systems, Digital Communication System, Electric Motor Drives

SKILLS

- **Languages:** Verilog, C++, Python, TypeScript, RTL
- **Tools:** Synopsys Design Compiler, Cadence Innovus, Virtuoso/AMS, ModelSim, Quartus Prime, Altium Designer, GitHub, Jira, Cadence Xcelium Logic Simulator
- **Devices:** Raspberry Pi, NVIDIA Jetson, Oscilloscope, Microscope, Soldering Station
- **Design Libraries:** Generic 90nm Process Design Kit, FreePDK 45, TSMC 65 nm process PDK, PyTorch, numpy

AWARDS

- **Dean's Honour List:**(2025, 2023, 2021)
- **McMaster Honour Award:**(2020)
- **Dr. Rudolf De Buda Scholarship:**(2025)

PROJECTS

Qual-Core Attention Accelerator | <https://github.com/JustinLinKK/quad-core-attention-accelerator>

Jan 2026 - Present

- RTL for four MAC cores full-chip implementing the QK normalize pipeline for Transformer inference
- Designed 4 cores architecture and finite state controller, synchronize cores' data via tree adder
- Executed full backend workflow: Synopsys Design Compiler (synthesis), Cadence Innovus (hierarchical PnR), Cadence Xcelium for gate-level VCD simulation
- Improve the throughput to 1GHz

2-D Systolic Array Design with CNN Implementation

Sep 2025 - Dec 2025

2-D Systolic Array Design with CNN Implementation

- RTL for a 2D systolic array supporting output- and weight-stationary CNN modes with local buffering
- Brought up the design on Cyclone IV GX FPGA, profiling power/throughput and tuning multi-tiling plus clock-gated datapaths, SIMD, Clock Gating and multi-tiling design
- Reached a 130MHz operating frequency with 340mW power in FPGA mapping, save 30% dynamic power with 80% sparsity model
- Prue CNN with 88% sparsity while maintain >90% precision in validation

FIR Filter Adder Optimization on ASIC Design | [FIR Filter Adder Optimization on ASIC Design](#)

Sep 2025 - Dec 2025

- Modeled ripple, carry-save, and Brent-Kung prefix adders in FIR datapaths; captured activity via ModelSim for synthesis estimates
- Generated Agilent 7 area/timing reports to justify the selected prefix + ripple-tail architecture for lower fanout and power

Wearable Cave Mapping System | [Wearable Cave Mapping System](#)

Jan 2025 - May 2025

- Led PCB design (power regulation, high-speed buses, MCU control) for a head-mounted 3D scanner
- Integrated IMU, LiDAR, and Depth Camera modules; validated 15 fps mapping over 3-hour battery runs
- Successfully perform stable 15fps process frequency with 3+ hour battery life within \$1000 budget

WORK EXPERIENCE

MediumAI

Jan 2023 - Present

Co-Founder & Software Engineer (MediumAI)

Canada

- Built production RAG LLM pipelines for multilingual medical transcription and multimodal documentation using Langgraph, vector database, and cloud architecture, improving transcription accuracy in Mandarin and Cantonese by 10%, reducing time to first token by 32%
- Led R&D planning and delivery using GitHub and Jira, coordinated a team of engineers to cut operating costs by 42% through a multi-agent, multi-modal approach
- Containerized services with Docker, orchestrated deployments with Kubernetes on Azure AKS, and maintained CI/CD pipelines, cutting deployment time by 40% and increasing system reliability
- Secured \$10,000+ investment at the initial stage through a technical demo and assisted expansion to over 15 clients in Toronto, enabling early market traction

Sun Yat-sen University

May 2023 - Aug 2023

Computer Science RA Intern

China

- Developed Python neural-network models to simulate physics phenomena and solve differential equations, maintain similar simulation accuracy and reducing computation time by 12%
- Processed and visualized large 3D scan datasets with Open3D and ParaView, creating interactive views that helped the research team identify key structural features